

JEFFREY ALLEN

+1 (720) 387-6266 | jeffallen2.71828@gmail.com | [linkedin.com/in/jeffrey-allen-9a0b82260](https://www.linkedin.com/in/jeffrey-allen-9a0b82260)

EDUCATION

University of Colorado Boulder

B.S. Electrical & Computer Engineering | GPA: 3.275

Boulder, CO

Aug. 2023 – May. 2026

University of California, Davis

Coursework in Engineering | GPA: 3.569

Davis, CA

Sept. 2022 – June 2023

TECHNICAL SKILLS

Power Topologies & Control	Buck, Boost, Inverting Buck-Boost, Cascaded Boost, DC-AC H-Bridge, LDOs, MPPT, PID, Loop Gain, Bode Plotting
Hardware & High Voltage	13.2 kV & 480 V/240 V Distribution, Custom Magnetics, Substation Panel Development, CV/CC Architectures, NEC Code
Tools & Software	LTspice, WEBENCH Power Designer, SIMetrix, Altium Designer, Fusion 360, VNA (Bode 100)
Embedded & Protocols	C/C++, ARM Cortex, USB-PD, EtherCAT, I ² C, SPI

EMPLOYMENT

The Boring Company

Electrical Engineering Intern

May 2025 – July 2025

Bastrop, TX

- Commissioned a 1.1 MVA substation, designing the 13.2 kV→480 V and 480 V→240 V split-phase distribution paths.
- Developed 480 V_{AC} electrical distribution panels integrating DC-DC converters, HRG, NGR, and VFDs.
- Produced comprehensive work instructions, structured BOMs, and single-line diagrams for field technician use.

PROJECT EXPERIENCE

100 W USB-PD Portable Variable Power Supply | Power Design Lead

Fall 2025 – Spring 2026

- Directed power flow architecture for a 100 W USB-PD bench supply, managing conversion and low-loss delivery from a 20 V/5 A USB-C contract.
- Designed a Power Rail Generation system using a TPS55340 boost and LM5116 inverting buck-boost pre-regulators feeding ±20 V LDOs with a π -filter input for optimal noise reduction.

Photovoltaics Power Lab | Power Electronics Engineer

Spring 2026

- Designed and built a complete 85 W PV power conditioning chain, transforming raw solar harvest into a synthesized 120 V_{AC}, 60 Hz grid-level output.
- Fabricated, gapped, and characterized custom power magnetics (N95 ferrite cores) for buck and cascaded boost converter stages.
- Implemented a custom C++ Perturb-and-Observe MPPT algorithm, achieving 97.76% converter efficiency at the Maximum Power Point.
- Validated control-to-output loop stability and phase margins using a Bode 100 Vector Network Analyzer.

CU Hyperloop TBM | Electrical & Software Team Member

Aug. 2024 – Sept. 2025

- Engineered a custom Fluids Board PCB with a modified relay-mosfet control loop driving 6 cooling and clay-dissolving pumps via an internal feedback loop.
- Designed environmental sensing PCBs (Navigation and Rear Gripper boards) to capture temperature, pressure, and methane telemetry.

PERSONAL ACHIEVEMENTS

National Eagle Scout Association

Dec. 2017 – Present

- Led a 40+ person volunteer team to renovate community property, managing logistics and execution from planning through completion.

Mountaineering: Denali Summit (20,320 ft)

June 2022

- Demonstrated endurance, risk management, and teamwork during a high-altitude multi-week expedition.